Real-Time Driver and Road Monitoring for Risk Prediction

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Project Proposal Report

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Declaration of The Candidate & Supervisor

I declare that this is my work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or institute of higher learning, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

29/08/2025

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Signature of the supervisor Date

Acknowledgment

This paper and the research behind it would not have been possible without the exceptional support of my supervisors, my Research supervisor Mrs. Manori Gamage Co-supervisor Mr. Nelum Chathuranga Their enthusiasm, knowledge, and exacting attention to detail have been an inspiration and kept my work on track. Besides my supervisors, I would like to thank the rest of my group members Senadheera S.C.C, Pinsara T.H.A.K, Samodha T W O for their encouragement, insightful comments, and every other support. And finally, thanks to my parents who always offer support and love.

Abstract

This research proposes a single-module fatigue detection system that integrates physiological, behavioural, and environmental indicators to continuously assess a driver's condition in real time. Physiological data, such as heart rate variability (HRV), will be captured using an ECG sensor to detect signs of physical strain. Behavioural indicators, including facial expressions and head pose, will be extracted from in-cabin camera footage using computer vision techniques. Environmental factors, such as time of day, weather, and driving context, will be incorporated to account for external influences on fatigue. The system employs a dynamic weighting mechanism, using fuzzy logic or a rule-based approach, to adjust the influence of each indicator based on context and sensor reliability. These weighted features will be processed by supervised machine learning models to compute a fatigue risk score. When the score exceeds a defined threshold, the system will generate real-time alerts, enabling proactive interventions. By combining multi-modal sensing with adaptive fusion, the proposed system aims to deliver accurate, context-aware fatigue prediction and significantly reduce fatigue-related accidents. The ultimate goal is to prevent accidents before they occur, ensuring the safety and well-being of both drivers and other road users.

Keywords: fatigue detection, heart rate variability, GSM modules, computer vision, multi-modal

Table of Contents

Table of Contents

Declaration Of The Supervisor..... 2

Acknowledgement..... 3

Table of Contents 5

[1] INTRODUCTION..... 6

1.1. Background & Literature Survey..... 6

1.2. Literature Survey..... 7

1.3. Research Gap 9

1.4. Research Problem.....10

[2] OBJECTIVES..... 11

2.1. Main Objective..... 11

2.2. Specific Objectives..... 12

[3] METHODOLOGY..... 13

3.1. Project Overview.....13

3.2. System Overview Diagram.....14

3.3. Individual Component..... 16

3.4. Tools And Technologies..... 19

[4] PROJECT REQUIREMENT..... 20

[5] WORK BREAKDOWN CHART.....23

[6] GANTT CHART..... 24

[7] BUDGET AND BUDGET JUSTIFICATION..... 25

[8] REFERENCES..... 26

INTRODUCTION

1.1 Background & Literature Survey

Road accidents are a serious and persistent problem in Sri Lanka, claiming thousands of lives and injuring many more each year. According to the Sri Lanka Police Traffic Division, fatigue and drowsy driving contribute significantly to highway and long-distance bus accidents, particularly on major routes such as the Southern Expressway and the A9 highway. Many professional drivers, including bus, truck, and taxi operators, work extended hours with insufficient rest, often under challenging road and weather conditions.

Sri Lanka's tropical climate, with its high temperatures and humidity, can further accelerate driver fatigue, especially during midday and early afternoon travel. In rural and less-developed regions, poor road infrastructure and the lack of designated rest stops exacerbate the problem. Additionally, cultural practices such as overnight travel for long-distance passenger transport increase the risk of fatigue-related accidents.

Advances in sensing and computation make it possible to address this problem. The increasing availability of smart sensors, IoT devices, and machine learning tools provides an opportunity to build intelligent monitoring systems that can track a driver's condition and the environment in real time. In particular, combining data from driver-worn sensors (e.g. ECG for heart rate), in-cabin cameras (for facial expressions and head pose), and external data (time, weather, location) could enable more accurate fatigue prediction. This project specifically targets early detection of driver fatigue through long-term monitoring. The goal is to design a single-module fatigue detection system that uses physiological, behavioural, and environmental data to continuously evaluate the driver's state. By processing these inputs with supervised machine learning, the system will compute a fatigue risk score or alert the driver before serious fatigue sets in. In this way, the system aims to enhance road safety by providing timely, context-aware warnings whenever the driver's fatigue level becomes critical.

1.1. Literature Survey

- **Thiffault and Bergeron (2003)** – Examined individual differences in driver fatigue during monotonous simulated driving. They found that personality traits such as sensation seeking (especially the "Experience Seeking" dimension) predicted steering variability, which served as a performance measure of fatigue. High-sensation-seeking individuals exhibited larger variations in steering and were more likely to report having fallen asleep at the wheel. This study highlights that personal factors and simple vehicle-based measures (steering wheel movements) can indicate fatigue.
- **Sahayadhas et al. (2012)** – Provided a comprehensive review of sensor-based driver drowsiness detection. They categorize detection methods into *vehicle-based* (e.g. lane deviation, steering wheel movement), *behavioral* (e.g. yawning, eye closure, facial gestures), and *physiological* measures (e.g. ECG, EEG, EOG signals). They emphasize that effective detection systems often combine multiple measures. Their review notes that drowsy driving remains a serious problem: about 100,000 U.S. crashes per year are directly attributed to driver sleepiness. This work underlines the need to fuse diverse data modalities to improve detection accuracy.
- **May and Baldwin (2009)** – Discuss driver fatigue causes and detection technologies. They stress that fatigue can be sub-categorized into *sleep-related* and *task-related* fatigue based on causal factors, and that distinguishing these types is important for designing effective countermeasures. They argue that without identifying causal factors, one-size-fits-all detection technologies may be less effective. This suggests that contextual factors (like circadian rhythm or time of day) should inform fatigue assessment.
- **Lal and Craig (2001)** – Critically reviewed psychophysiological aspects of driver fatigue (Biological Psychology, 2001). They summarize how physiological signals (e.g. brain EEG patterns, heart rate variability, skin conductance) change with fatigue and can be used for detection. While we do not quote their text directly, this review highlights that physiological monitoring (like ECG-derived HRV) provides objective cues to fatigue.
- **Rosekind et al. (1994)** – Examined fatigue in operational settings (aviation) in *Human Factors*. They provide examples of how fatigue manifests in high-stakes environments, implying that fatigue monitoring can improve safety in various domains.
- **Horne and Reyner (1995)** – Studied sleep-related vehicle accidents (BMJ 1995). Their surveys in England found that sleep-related crashes typically occur in three peak periods (around 2:00–6:00 am and 2:00–4:00 pm) and account for a large fraction of all accidents on motorways and monotonous roads. About half of these drivers were

young men under 30. This work shows that environmental context (time of day, road type) critically influences fatigue risk.

These studies collectively underscore the multifaceted nature of driver fatigue and its detection: personal and vehicular measures can indicate fatigue, sensor-based techniques cover multiple modalities, and contextual factors (circadian effects, environmental conditions) play a large role. The literature suggests that no single indicator suffices; instead, robust detection likely requires fusing diverse data streams with awareness of context.

1.2. Research Gap

Our research endeavors to elucidate a discernible research lacuna, aiming to bridge the void between the established knowledge and the unexplored facets within the domain of transmitting information to stakeholders and monitoring driver movements upon signal reception. This pursuit is indispensable, not solely for the progression of scholarly discourse, but also for the pragmatic resolution of challenges and the advancement of objectives pertaining to the efficient dissemination of information to drivers.

Table 1: Comparison with existing system

Reference	Limited integration of physiological, behavioral & environmental indicators	Lack of solutions tailored for Sri Lankan driving conditions	Absence of adaptive weighting mechanisms for sensor reliability	High cost / low affordability for local fleets
[1] Thiffault & Bergeron	Yes	Yes	Yes	No
[2] Sahayadhas et al.	Yes	Yes	Yes	Yes
[3] May & Baldwin	Yes	Yes	Yes	No
[4] Lal & Craig	Yes	Yes	Yes	Yes
[5] Rosekind et al.	Yes	Yes	Yes	No
[6] Horne & Reyner	Yes	Yes	Yes	No
Proposed System	No	No	No	No

In this table:

- "Yes" indicates that the paper does not address the specified challenge.
- "No" indicates that the paper addresses the specified challenge.

1.3. Research Problem

Driver fatigue is a complex and multifaceted issue that significantly contributes to road accidents worldwide, and its impact is particularly severe in contexts such as Sri Lanka, where long-distance travel, overnight driving, and challenging environmental conditions are common. Fatigue impairs a driver's cognitive and motor functions, slowing reaction times, reducing alertness, and increasing the likelihood of critical errors. Despite advances in vehicle safety technology, many existing fatigue detection systems are limited in scope, focusing on either behavioral indicators (e.g., eye closure, head movement) or physiological signals (e.g., heart rate), often neglecting the critical influence of environmental factors such as time of day, weather, and road context.

Moreover, these systems typically rely on static weighting methods for data interpretation, meaning they cannot adapt to changing conditions or sensor reliability. For example, a camera-based system may lose accuracy in low-light situations, yet its detection logic remains unchanged, leading to false negatives or positives. Additionally, many commercially available solutions are expensive, require multiple hardware modules, and are designed for developed countries with stable road infrastructure, making them unsuitable for widespread use in Sri Lanka.

Therefore, the central research problem is:

How can we design a single, affordable, and context-aware fatigue detection system that integrates physiological, behavioral, and environmental indicators, dynamically adapts to sensor reliability and changing driving conditions, and delivers accurate real-time fatigue risk assessments and alerts to prevent accidents?

This problem encapsulates the technical, economic, and contextual challenges of creating a practical solution that not only leverages multi-modal sensing and machine learning but is also tailored to the realities of local driving environments.

2. OBJECTIVES

2.1 Main Objective

The main objective of this system is to design and develop an integrated, real-time, single-module fatigue detection platform capable of accurately identifying and predicting driver fatigue by combining physiological, behavioral, and environmental monitoring into one cohesive, adaptive solution.

This objective centers on creating a comprehensive safety mechanism that continuously evaluates the driver's condition using diverse and complementary data sources:

Physiological monitoring through heart rate variability (HRV) analysis derived from ECG signals captured by the AD8232 sensor, enabling early detection of stress and physical strain.

Behavioral analysis through computer vision techniques applied to real-time in-cabin video, detecting critical fatigue cues such as prolonged eye closure, yawning, and abnormal head pose angles in three-dimensional space (yaw, pitch, roll).

Environmental assessment through sensor-based measurements of temperature, humidity, ambient light, rainfall, and driving context like urban, highway, or overnight conditions, allowing the system to account for external factors that significantly influence fatigue onset.

By fusing these inputs through a dynamic weighting mechanism using fuzzy logic or rule-based adaptation the system aims to provide a fatigue risk score that reflects the driver's true condition in context, rather than relying on a single indicator. This score is used to trigger timely alerts, ensuring that the driver is warned before fatigue reaches a dangerous level.

Ultimately, the primary goal is to prevent fatigue-related accidents by delivering an affordable, reliable, and easily deployable solution that enhances driver awareness, protects passenger safety, and contributes to overall road safety in diverse operating conditions, including the challenging climatic and infrastructural realities of Sri Lanka.

2.2 Specific Objectives

To achieve the main objective, the specific objectives that need to be attained are in the proposed system as follows,

- Physiological Monitoring**:** Acquire heart rate signals using an ECG sensor (e.g. AD8232 module) and extract heart rate variability (HRV) features that correlate with fatigue.

- Behavioral Analysi**s:** Use an in-cabin camera to detect fatigue-related facial expressions (e.g. yawning, eye closure) and estimate head pose (e.g. nodding) using computer vision techniques (e.g. OpenCV for head pose detection).

- Environmental Context**:** Gather contextual data such as current time of day, weather conditions, and road type (urban vs highway) from appropriate sources (e.g. GPS, APIs)

- Adaptive Fusion**:** Implement a fuzzy logic or rule-based engine to assign dynamic weights (X, Y, Z) to physiological, behavioral, and environmental indicators based on context or sensor reliability.

- Machine Learning Model**:** Train a supervised machine learning model (e.g. a classification or regression model) on labeled data (for instance, using the Facial Expression of Fatigue dataset and collected heart rate data) to predict the driver's fatigue level from the fused features.

- Real-Time Alerting**:** Integrate the components into a real-time pipeline that continually computes a fatigue risk score. Generate on-screen or audible alerts when the score exceeds a safety threshold, thereby enabling proactive intervention.

3. METHODOLOGY

3.1 Project Overview

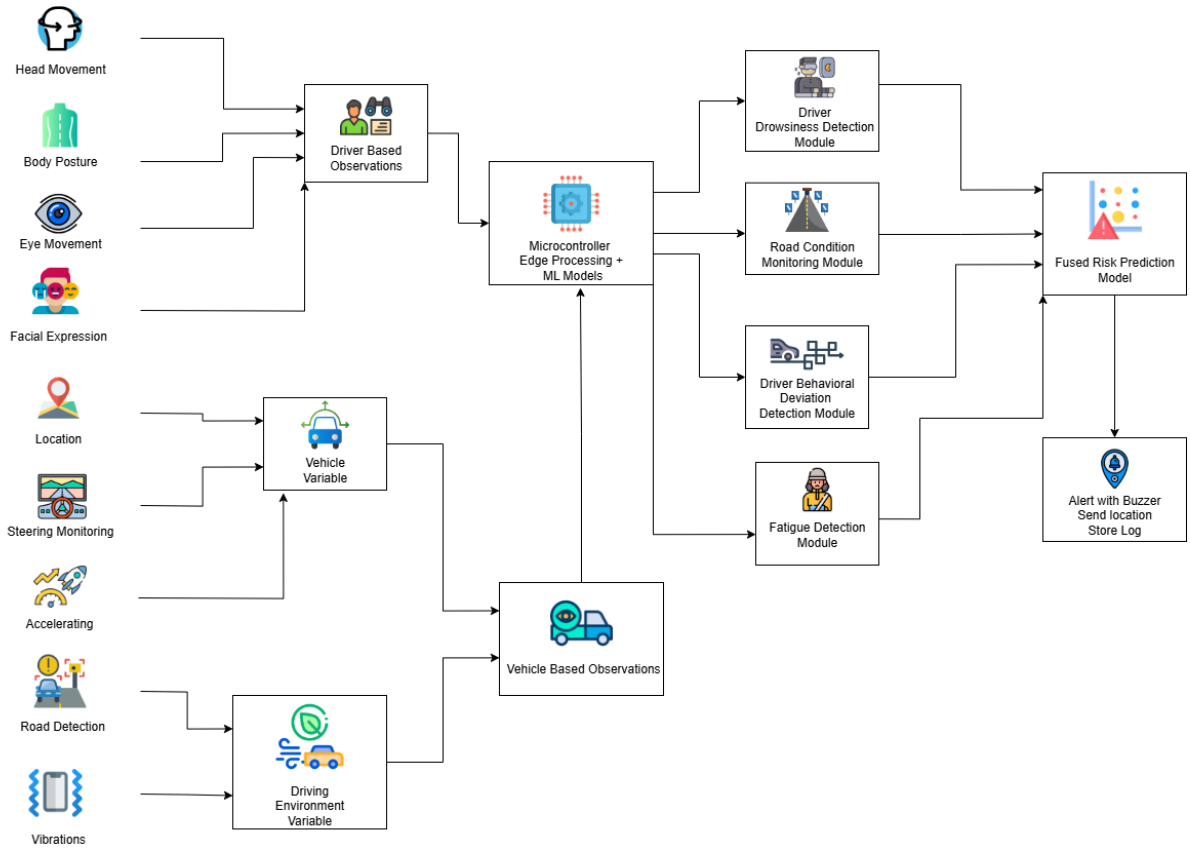

Figure 2: Project Overview Diagram

3.2 System Overview Diagram

The Driver Drowsiness Risk Evaluation System is a comprehensive, multi-module safety platform designed to assess and mitigate driver drowsiness and fatigue in real time. It integrates four distinct yet interconnected modules—Driver Drowsiness Detection, Road Condition Monitoring, Driving Pattern Analysis, and Fatigue Detection—into a unified architecture powered by microcontroller-based edge processing and machine learning models.

The system continuously collects data from a range of sources, including in-cabin cameras, physiological sensors, vehicle telemetry, and environmental monitoring devices. These inputs are processed locally on the microcontroller to ensure low-latency analysis without heavy reliance on cloud computing, making the system suitable for deployment in both urban and rural transport scenarios.

Driver Drowsiness Detection module estimates the driver's drowsiness risk by combining physical factors (e.g., blood pressure, heart rate changes) with environmental factors (e.g., temperature, ambient light, and noise levels). It calculates a composite drowsiness score, and when thresholds are exceeded, it triggers an alert system to warn the driver. Unlike traditional vision-only systems, it incorporates multi-sensor fusion to improve accuracy in varying lighting and weather conditions.

Road Condition Monitoring module Designed to enhance road safety, this module utilizes on-vehicle camera feeds, weather sensors, and vehicle speed data to detect road signs (especially speed limits), assess road surface conditions (dry, wet, slippery), and monitor adherence to speed regulations. When hazards are detected—such as wet roads at high speeds or ignored speed limits—the system issues real-time audio-visual warnings to the driver.

Driver Behavioral Deviation Detection Module focuses on personalized driver behavior analysis. It learns each driver's typical driving patterns, including speed control, acceleration/braking habits, and steering movements. After a learning phase, the system identifies deviations that may indicate unsafe or distracted driving. Drivers can also select their experience level (beginner, intermediate, expert) to adjust alert sensitivity during the learning period. Over time, the system transitions to a fully personalized risk model.

Fatigue Detection module integrates physiological monitoring using ECG-based HRV analysis, behavioral monitoring using facial expressions and head pose via computer vision and environmental sensing light, temperature, humidity, and weather into a single adaptive model. Data from all sources are fused using fuzzy logic or rule-based dynamic weighting, producing a real-time fatigue risk score. When this score surpasses a critical threshold, the system activates a tiered alert mechanism visual, audio ensuring the driver is warned before fatigue reaches dangerous levels.

Integrated Fused Risk Prediction Model

The outputs of all four modules are processed by a central fusion engine that generates an overall Drowsiness and Fatigue Risk Index. This risk index considers both immediate threats like detected drowsiness, poor road conditions and long-term fatigue patterns. If the index exceeds the set safety limit the system Triggers an alert with a buzzer and visual cues. Stores a detailed log for post-event analysis. This modular yet integrated approach ensures redundancy, context awareness, and high detection accuracy, significantly improving road safety by addressing multiple drowsiness and fatigue factors simultaneously.

3.3 Individual Component

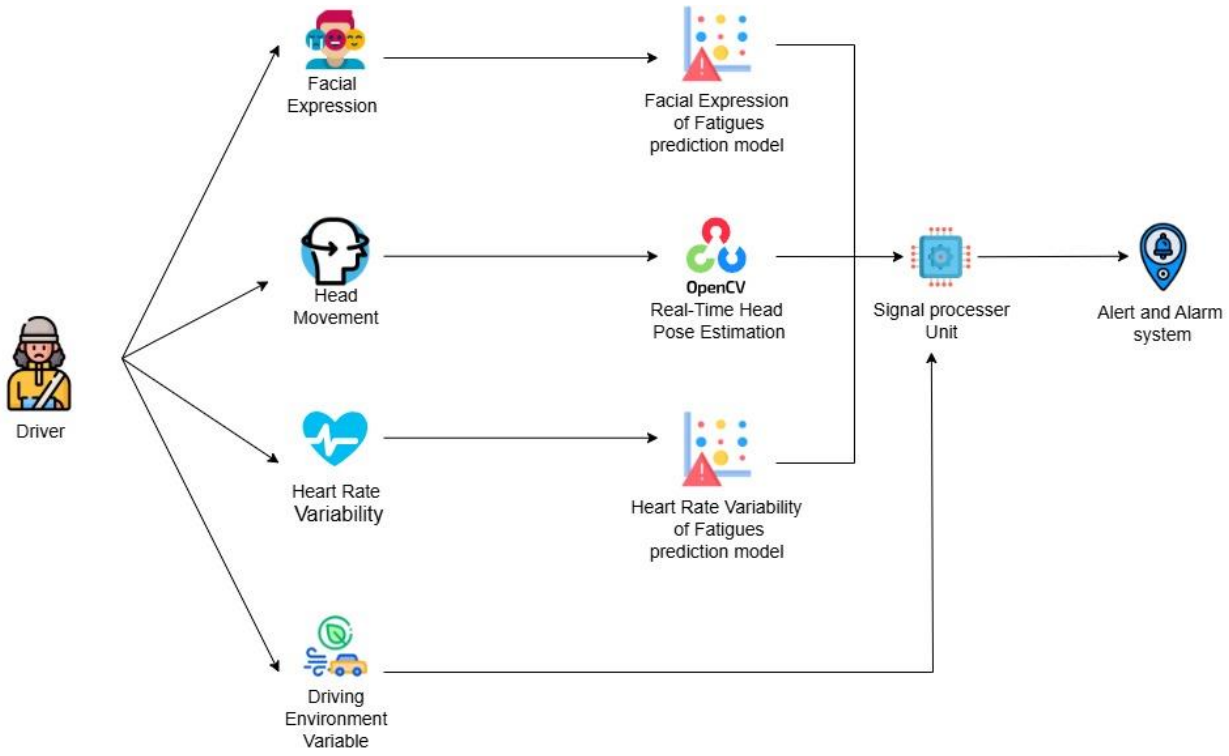

Figure 3: Project Overview Diagram

3.3.1 The flow of the system diagram – Individual Component

The proposed system operates through a structured sequence of interconnected components designed to provide continuous and accurate detection of driver fatigue. At its core, the system integrates three primary data streams behavioral cues captured via a camera, physiological signals obtained through an AD8232 ECG module pulse heart rate monitor, and environmental conditions measured through dedicated sensors.

The camera module is mounted to maintain a clear view of the driver's face and upper body. It continuously captures real-time video during vehicle operation. This video feed is processed using the OpenCV Python library to determine the driver's head pose in three dimensions—yaw, pitch, and roll representing the X, Y, and Z axes. Deviations from normal head orientation, such as forward nodding or lateral tilting, are recognized as potential signs of fatigue. Concurrently, the same camera feed is analyzed using a Convolutional Neural Network (CNN) trained on the Facial Expression of Fatigue (FEF) dataset, enabling the detection of subtle and overt facial fatigue indicators such as prolonged eyelid closure, yawning, or reduced blink frequency.

The physiological monitoring component employs the AD8232 ECG module to collect the driver's heart signal data. From these signals, the system calculates heart rate variability (HRV), a clinically recognized measure of fatigue and stress levels. Data acquisition is time-synchronized with the camera feed, allowing for cohesive multi-modal analysis. Raw ECG signals are first filtered and pre-processed using an Arduino microcontroller, which transmits the cleaned data to the processing unit for feature extraction. A supervised machine learning model, trained on labelled HRV datasets, predicts the driver's fatigue likelihood from physiological patterns.

The environmental monitoring subsystem integrates multiple sensors to capture external and in-cabin conditions that influence fatigue. BME280 sensor measures ambient temperature, humidity, and barometric pressure, enabling the computation of a heat index and monitoring weather-related fatigue factors. BH1750 digital light sensor measures ambient light levels in lux, allowing the system to identify low-light conditions such as night time driving or poor weather visibility. Weather API detects precipitation, which can further influence fatigue risk due to increased driving stress. GPS module NEO-6M determines time, speed, and driving context like urban or highway, which are critical in assessing fatigue patterns, especially during prolonged high-speed travel or overnight journeys. These environmental variables are continuously recorded and processed to provide contextual awareness.

All three subsystems feed their outputs into the data fusion and decision-making layer. This stage employs a fuzzy logic or rule-based adaptive weighting system to dynamically adjust the influence of each data category physiological (X), behavioral (Y), and environmental (Z) based on real-time sensor reliability and driving context. For lets assume, if the BH1750

detects very low light and camera quality degrades, the system automatically reduces the behavioral weight and increases reliance on physiological and environmental inputs.

The fused feature set is processed by the central fatigue assessment algorithm, which calculates a fatigue risk score on a continuous scale between 0 - 1. When the score exceeds a predefined threshold, the alert module is triggered. The alert system can issue tiered warnings a soft visual cue for moderate risk, and a strong audible or haptic alert for high-risk conditions sustained over a set period.

By integrating camera-based behavior analysis, ECG-derived HRV monitoring, and environmental sensing through BME280, BH1750, rain, and GPS modules, this system provides a highly accurate, adaptive, and context-aware fatigue detection capability. Its single-module design ensures compactness, ease of integration into existing vehicles, and cost-effectiveness, making it suitable for both commercial and personal transport applications while aiming to prevent fatigue-related accidents before they occur.

3.4 Tools And Technologies

Table 4: Tools and Technologies

Tool and Technologies	Explanation
Arduino Boards	Interface with sensors, preprocess signals, and send data to main processing unit.
AD8232 ECG Sensor Module	Measure heart activity and extract Heart Rate Variability (HRV) to detect physical fatigue.
BME280 Sensor	Measure temperature, humidity, and barometric pressure for environmental condition tracking.
BH1750 Light Sensor	Detect ambient light levels to assess day/night and visibility conditions.
GPS Module (NEO-6M)	Provide time, speed, and driving context (urban vs. highway).
Arduino IDE	The Arduino IDE (Integrated Development Environment) is a software application used to write, compile, and upload code to Arduino boards. It's a crucial tool for anyone working with Arduino, allowing users to create programs (sketches) that control the board's behavior.
Python	Main programming language for system control, data analysis, and ML model execution
Camera Module	Capture real-time video of driver's face for head pose and facial expression analysis.
TensorFlow With Keras	Build and train CNN models for facial expression classification.
OpenCV	Real-time image processing for head pose estimation.

2 PROJECT REQUIREMENTS

4.1 Functional Requirements

[1]. Driver Behavior Monitoring

- The system shall capture real time in-cabin video using a camera to monitor the driver's facial expressions and head pose.
- The system shall use the OpenCV library to calculate head pose in three dimensions of yaw, pitch and roll.
- The system shall detect fatigue-related facial expressions like yawning, prolonged and eye closure using a CNN trained on the *Facial Expression of Fatigue (FEF)* dataset.

[2]. Physiological Signal Monitoring

- The system shall measure heart rate signals using the AD8232 ECG sensor.
- The system shall compute Heart Rate Variability (HRV) from the ECG data to assess physical strain and fatigue.
- The system shall preprocess raw ECG signals using an Arduino or ESP32 before analysis.

[3]. Environmental Monitoring

- The system shall measure ambient temperature, humidity, and barometric pressure using the BME280 sensor.
- The system shall measure ambient light intensity using the BH1750 sensor to identify day/night conditions.
- The system shall detect rainfall using a weather API.
- The system shall capture time, speed, and driving context via a GPS module.

[4]. Data Fusion and Analysis

- The system shall integrate physiological, behavioral, and environmental data into a unified fatigue assessment.
- The system shall dynamically adjust the weighting of each data category (X, Y, Z) using fuzzy logic or rule-based adaptation based on sensor reliability and driving context.
- The system shall calculate a real-time fatigue risk score.

[5]. Alerting and Intervention

- The system shall trigger visual, audible, or haptic alerts when the fatigue risk score exceeds a defined threshold.
- The system shall provide tiered alerts Soft Alert for moderate risk.
- Strong Alert for high risk sustained over a certain duration.

[6]. Data Logging

- The system shall log time-stamped sensor readings, calculated scores, and alert events for post-analysis

4.1 Non- Functional Requirements

[1]. Performance

- The system shall process incoming data and update the fatigue risk score in less than 1 second for real-time responsiveness.
- The system shall maintain consistent performance under variable lighting, temperature, and network conditions.

[2]. Accuracy

- Behavioral detection models shall achieve ≥90% accuracy in classifying fatigue-related facial expressions.
- HRV-based physiological analysis shall achieve ≥85% accuracy in distinguishing normal and fatigued states.

[3]. Reliability

- The system shall maintain continuous monitoring without crashes or interruptions during the entire driving session.
- The system shall degrade gracefully if one or more sensors fail, adjusting weights to use remaining data sources.

[4]. Scalability

- The design shall support easy integration of additional sensors or improved models without major hardware changes.

[5]. Usability

- The alerts shall be clear, non-intrusive, and easily understood by the driver without causing distraction.
- The system shall require minimal setup and calibration before use.

[6]. Portability

- The module shall be compact and lightweight for installation in different vehicle types.

[7]. Cost-effectiveness

- The system shall be affordable for deployment in public transport fleets, logistics vehicles, and private cars.

3 Work Breakdown Chart

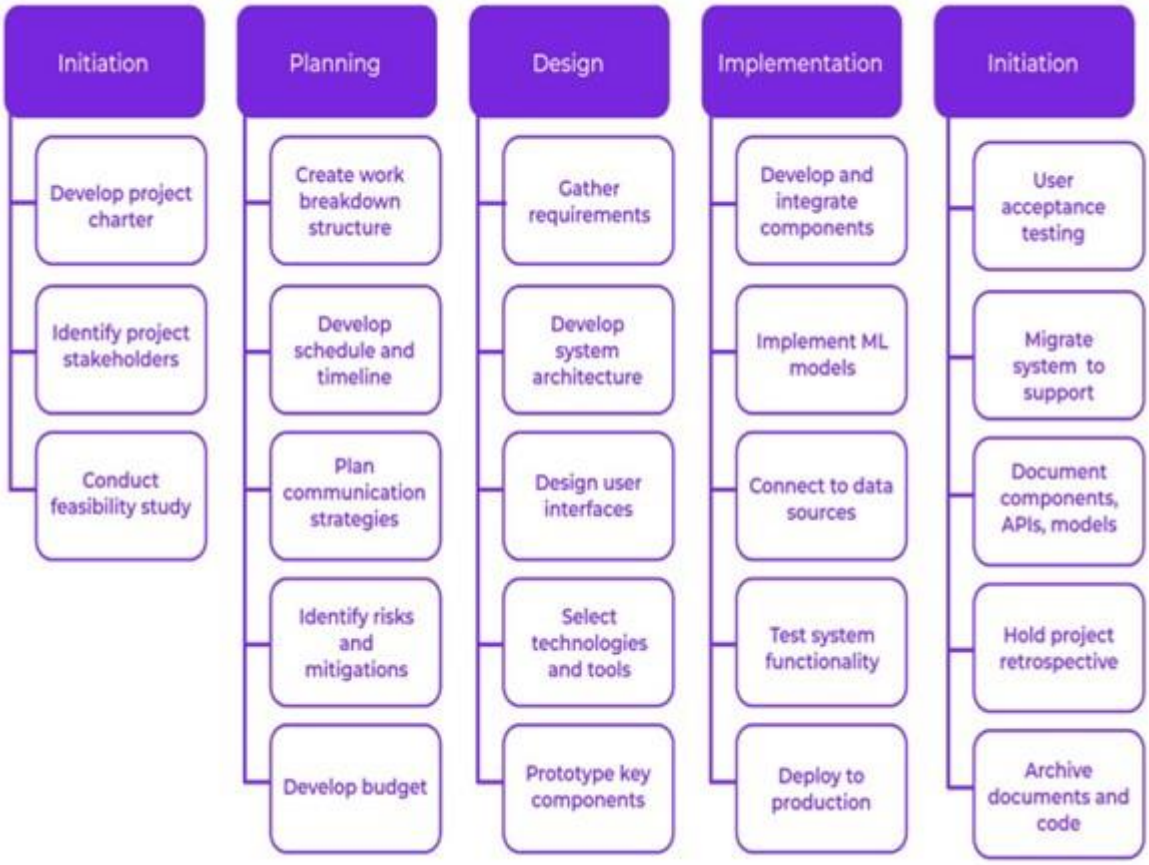

Figure 5: Work Breakdown Chart

4 Gantt Chart

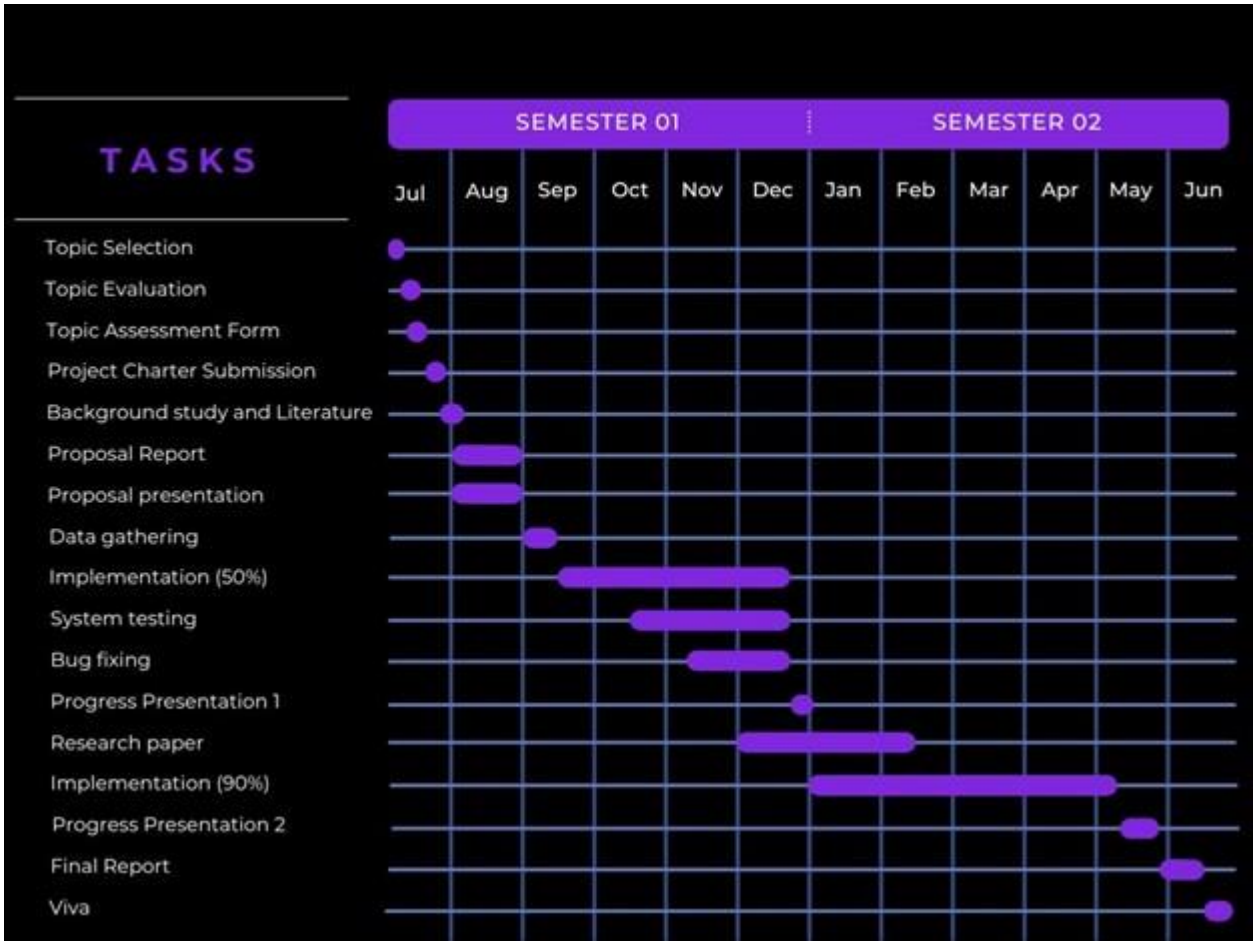

Figure 6: Gantt Chart

4. BUDGET AND BUDGET JUSTIFICATION

Determining the budget for the development and implementation of the driver and road Detection System for requires a comprehensive analysis of various factors, including the scope of the project, technology requirements, personnel costs, infrastructure expenses, and ongoing operational expenses. While providing an exact budget without detailed information is challenging, I can outline some key cost considerations

Table 7: Budget and Budget Justification

Component	Amount (LKR)
Arduino Boards	2,850
AD8232 ECG Sensor Module	1,450
BME280 Sensor	950
BH1750 Light Sensor	1,650
GPS Module (NEO-6M)	1,510
Alert Module	4,000
Camera Module	3,550
Total	15, 990

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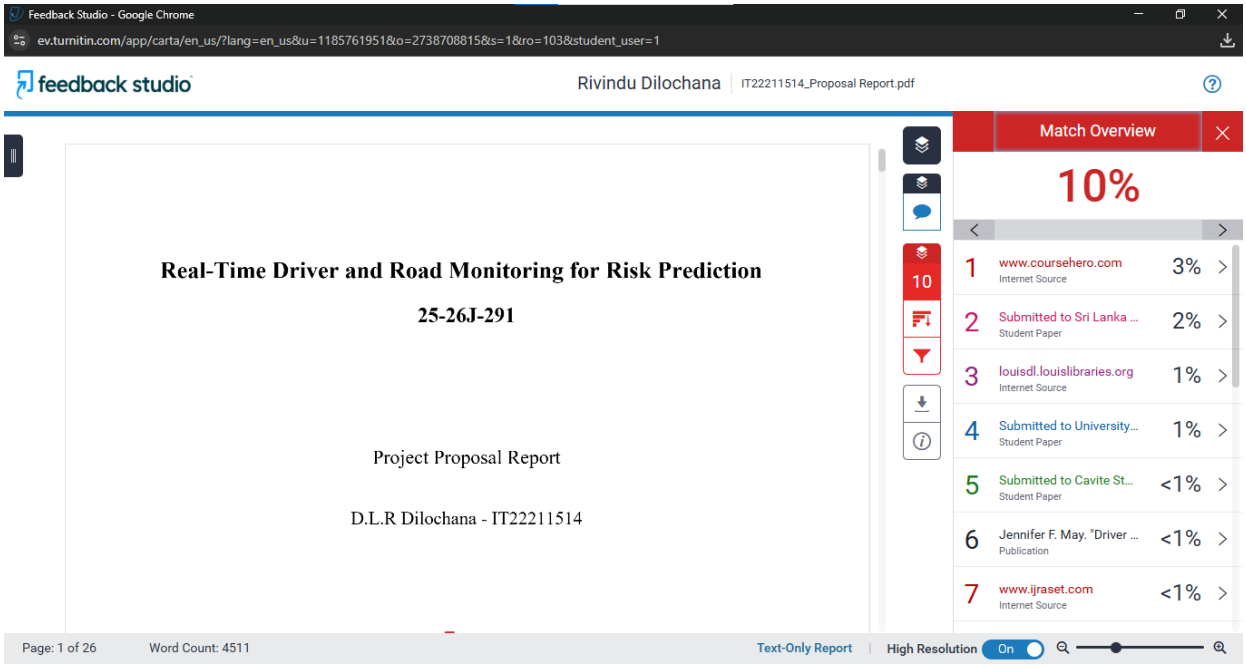
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